

Michael Hasson, Ph.D.

Stanford University

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Summary

As a Neukom Fellow, I currently focus on quantifying the risks that climate change poses to humans and human infrastructure. In my past research, I have focused on unraveling how physical processes shape the surface of the Earth and how those processes are represented in Earth's stratigraphic record. I have worked at a range of scales, from individual grains of sand to the entire globe, and use a combination of remote sensing, machine learning, and fieldwork to understand Earth's past and present.

Education

Stanford Department of Earth & Planetary Sciences, Ph.D. June 2026

- Ph.D. Candidate 2022-2026
Advisor: Dr. Mathieu Lapôtre

- Ph.D. Student 2020-2022

Colorado College Department of Geology, B.A., Distinction in Geology 2019

Professional Experience

Neukom Institute for Computational Science, Dartmouth College Hanover, NH
Neukom Postdoctoral Fellow September 2026-present

- Co-mentors: Justin Strauss and Justin Mankin.

Orbital Sidekick San Francisco, CA
Data analytics intern Sept.-Dec. 2024

- Designed and implemented workflow to prevent hydrocarbon leaks by monitoring oil and gas assets in the continental United States.
- Developed sensor-agnostic change detection algorithm combining multi- and hyperspectral spaceborne sensors.
- Projected revenue >\$2M annually with current contracts.

Colorado College Colorado Springs, CO
Paraprofessional 2019-2020

- Led labs and field trips for upper and lower-division undergraduate geology courses of up to 25 students.

TRC Consulting Concord, CA
Geological intern May-Aug. 2016

- Analyzed and documented environmental contaminant data collected from various in-progress remediation sites.

Technical

- **Machine learning:** TensorFlow, PyTorch, scikit-learn, xgboost
- **Geospatial:** Google Earth Engine, QGIS, ArcGIS, UAV stereophotogrammetry, LASTools, CloudCompare
- **Data management, analysis, and processing:** Python, Matlab, AWS
- **Other:** Experience with large (>50 TB) geospatial datasets (using local, cloud, and cluster computing resources), scanning electron microscopy, Adobe suite

Publications

Hasson, M., Marvin, M.C., and Lapôtre, M.G.A. Zircon transport histories revealed by automated classification of scanning electron microscope images. *In revision, Journal of Sedimentary Research.*

Hasson, M., Ielpi, A., Lapôtre, M.G.A. Transient and long-term dynamics of unvegetated meanders from high-resolution topographic change detection. *Submitted, Earth and Planetary Science Letters.*

Hasson, M., Finotello, A., Ielpi, A., Lapôtre, M.G.A. (2025). Vegetation alters the trajectory of river meander bends. *Science*, doi:10.1126/science.adv4939.

Marvin, M.C., **Hasson, M.,** Colicci, V., Abubo, R., and Lapôtre, M.G.A. (2025). Detrital zircon microtextures as a powerful new tool to interpret the Precambrian sedimentary record. *Geology*, doi:10.1130/G53712.1.

Hasson, M., Marvin, M.C., and Lapôtre, M.G.A. (2024). Determining depositional environments in sand and sandstones using quartz microtextures and deep learning. *Proceedings of the National Academy of Sciences*, doi: 10.1073/pnas.2407655121.

Hasson, M., Marvin, M.C., Gunn, A.L., Lapôtre, M.G.A., and Ielpi, A. (2023). A depositional model for meandering rivers without land plants. *Sedimentology*, doi:10.1111/sed.13121.

Myrow, P.M., **Hasson, M.,** Taylor, J.F., Tarhan, L., Fike, D.A., Ramirez, G., Fowlkes, G., Popov, L.E., Liu, H., and Chen, J. (2023a). Revised Paleozoic depositional history of the central Rocky Mountains (Utah and Colorado): *Sedimentary Geology*, doi:10.1016/j.sedgeo.2023.106373.

Myrow, P.M., **Hasson, M.,** Taylor, J.F., Tarhan, L.G., Ramirez, G., Fowlkes, G., and Chen, J. (2023b). Structural control of Cambrian paleotopography and patterns of transgression in western Laurentia: *Geology*, doi:10.1130/G51055.1.

Invited Talks

- Golden Gate STEM Fair, Randall Museum, March, 2026
- AGU Fall Meeting 2025: Vegetation Changes the Trajectory of River Bends (EP24C), December, 2025
- Paul Myrow Retirement Symposium, November, 2025
- Caltech GeoClub Seminar, May, 2025
- Coastal Morphodynamics and Machine Learning (COMOMoLA) seminar, February, 2025

In the News

Vegetation Changes the Trajectory of River Bends:

- [Science Perspective by J. Pizzuto](#), [Stanford News](#), [Radio France \(French\)](#), [Science Daily](#), [Environmental News Network](#), [The Hindu](#)

SandAI: An AI tool to decipher the history of sand grains:

- [Science Friday](#), [Stanford Report](#), [Radio Canada \(French\)](#), [Imaging and Machine Vision Europe](#)

Non-academic Publications

Hasson, M., 2018, Geology of the Sierra Nevada in *Discovering the John Muir Trail* by Damon Corso (Falcon Guides, 2018).

Honors and Awards

- Outstanding Graduate Award, Stanford University, 2026
- Achievement Rewards for College Scientists (ARCS) Scholar, 2025
- NSF Student Travel Grant, 2024
- Stanford Centennial Teaching Award, 2022
- NSF Graduate Research Fellowship Program Honorable Mention, 2022
- McGee/Levorsen Graduate Student Research Grant, Stanford University, 2022
- National Center for Airborne Laser Mapping (NCALM) Seed Proposal Winner, 2021
- McGee/Levorsen Graduate Student Research Grant, Stanford University, 2020
- Estwing Outstanding Senior Geologist Award, Colorado College Dept. of Geology, 2019
- Patricia J. Buster Research Scholarship, Colorado College, 2019

Teaching Experience

Guest lecturer:

- U.C. Berkeley: Civil Engineering 203N: *Surface Hydrology*, 2023
- Stanford: Earth 1C: *Know Your Planet – Science in the Field*, 2022

Teaching assistant:

- Earth & Planetary Sciences 3: *Earth and Planetary Processes and Mechanics*, 2023
- Earth & Planetary Sciences 224: *Rivers: The Arteries of Earth's Continents*, 2022

Outreach and Community Engagement

- Science and Education Lead, Surfrider Foundation, Santa Cruz Chapter, 2025-present.
 - Founded coastal science and conservation speaker series.
 - Obtained sponsorship from Patagonia.
- Science consultant with BBC Studios Science Unit for TV program entitled “Green Earth,” 2022.

Other

Long-distance hiking:

- Pacific Crest Trail thru-hike attempt, 2014 (approx. 1,300 miles).
- Pacific Crest Trail full thru-hike, 2015 (2,650 miles).

Rock climbing:

- Sport redpoint to 5.13b, boulder to V11, trad redpoint to .12b/c.

Expedition grants funded (Ritt Kellogg Memorial Fund, Colorado College):

- Alpine climbing, Wind River Range, Wyoming, 2016
- Sierra High Route thru-hike, California, 2017

- Sailboat-accessed rock climbing, British Columbia, Canada (as vessel captain and climber), 2018